The Heterogeneous Impacts of Job Displacement: Evidence from Canadian Job Separation Records*

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Abstract

We use Canadian administrative data linked to job separation records to understand the anatomy of mass layoffs and job-loss outcomes. Conventional identification approaches mask substantial cross-sectional heterogeneity in mass-layoff separations, mixing layoffs with quits, and employer identifier changes without job losses. Estimates from this homogenized sample severely understate both earnings losses and the role of employer effects in explaining these losses for laid-off workers. Layoffs incur larger and more persistent losses than quits, with moves to lower-paying employers explaining the long-term gap. These findings underscore the importance of distinguishing separation reasons in quantitative models of job dissolution and policy evaluation.

Keywords: Unemployment, earnings losses, layoffs vs quits
JEL Codes: E24, E32, J31, J65

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1 Introduction

Mass layoffs are events of central importance. They displace large numbers of workers from distressed or restructuring employers, many of whom face large and persistent earnings losses. A large literature analyzes the consequences of job separations during mass layoffs (Jacobson, LaLonde, and Sullivan 1993; Couch and Placzek 2010; Davis and von Wachter 2011; Bertheau et al. 2023, among others). In this paper, we use Canadian administrative data to study mass layoffs both as events of interest in their own right and as a tool for understanding the consequences of layoffs and other forms of separations. We make three key contributions. First, we revisit conventional approaches to identifying mass layoffs and quantifying their outcomes. We show that separations identified during mass layoffs exhibit substantial heterogeneity, both in whether they constitute actual job losses and, among actual separations, in the reason for separation. Second, we show that average job displacement outcomes mask large differences in the magnitude and persistence of earnings losses between layoffs and quits. Finally, we uncover the role of employers and job transitions in explaining the heterogeneity of earnings losses upon separations.

Studying mass layoffs requires employer–employee matched data to detect large employer contractions and track worker outcomes within the organization. In such data, separations are typically inferred from changes in employer identifiers. While this allows researchers to identify contractions, a common limitation is the lack of detailed information on separations. This presents two issues. First, employer identifier changes may reflect reorganizations rather than job losses. Second, even after removing spurious separations, the sample can include quits and separations arising for other reasons. Both complications may lead conventional approaches to understate the severity of layoff outcomes, and importantly, obscure key differences in the sources of those losses.

To evaluate the quantitative relevance of these issues for understanding mass layoffs, we use the Canadian Employer-Employee Dynamics Database (CEEDD) merged with Records of Employment (ROE), which employers must issue when earnings are interrupted. The ROE forms contain employment start and end dates, as well as separation reasons. To enable comparisons with

existing work, our benchmark focuses on long-tenure workers separating during mass layoffs in 2008–2010, as in Lachowska, Mas, and Woodbury (2020). We highlight three main findings.

First, existing mass-layoff identification strategies are predisposed to mixing in spurious separations. In our sample, 48.7% of identified separations have no ROE. We show that these missing ROEs (i) are more likely to exhibit concentrated worker flows between employers, where large groups of workers share the same origin and destination employers, (ii) deliver earnings paths that resemble those of stayers, and (iii) lead to a very low incidence of employment insurance (EI) receipt. These findings suggest that many employer identifier changes reflect reorganizations rather than actual job losses. Using the ROE, we evaluate the commonly-used approach of addressing concentrated flows by excluding flows that exceed a given *share* of the origin or destination employer's workforce. In our data, lowering the exclusion threshold from the commonly-used value of 80% to 50% removes many missing-ROE observations with limited additional exclusions of actual layoffs and quits, offering guidance for researchers without separation-reason data.

Second, we show that, even within mass-layoff separations *with* ROEs, only less than half are actual layoffs, while the rest are quits and idiosyncratic separations. Importantly, we document that accounting for the reason for separation is crucial to quantify both the level and persistence of earnings losses and the role of employer-specific pay premium (á la Abowd, Kramarz, and Margolis 1999, henceforth AKM) in explaining these losses. In terms of earnings losses, while short-term (1-year) earnings losses are 29 log points for all separators (including layoffs, quits, missing ROEs and idiosyncratic separations), they are 69 log points for layoffs and 25 log points for quits. Long-term (6-year) earnings losses are 11, 21, and 5 log points, respectively. Even if we assume away missing ROEs and idiosyncratic separations by pooling *only* layoffs and quits, which is ideal but highly unlikely for databases without ROE information, average losses still understate the level and persistence of the losses of laid-off workers. In terms of the role of employer premium, the lost employer premium accounts for only around 18% of short-term and 8% of long-term earnings losses for all separators. However, for layoffs, employer premium plays a much larger role, accounting for 28% of short-term and 58% of long-term losses. Thus, transitions to lower-paying employers are a

meaningful driver of laid-off workers' earnings losses, especially in the long run.

Third, focusing only on layoffs and quits in mass layoffs, we find that the sources of earnings losses largely vary depending on the reason for separation. Specifically, employer-premium losses are larger and more persistent for layoffs than for quits. While employer-premium loss is important to account for both short-term and long-term earnings losses for layoffs, it does not play any role for quits. Laid-off workers are more likely to make downward movements along the employer-premium ladder. Among downward movers, however, quitters are compensated with better employer-worker matches on average, implying that some accept lower pay premiums in exchange for better matches. Allowing for endogenous mobility using the approach of Bonhomme, Lamadon, and Manresa (2019) (henceforth BLM) preserves the large gap in these outcomes between layoffs and quits.

Our analysis complements Flaaen, Shapiro, and Sorkin (2019), who link survey data with administrative data to study the differences in earnings losses by reported reason for separation. Using fully administrative data, we characterize a detailed cross-sectional anatomy of mass-layoff separations, examining their composition, the prevalence of spurious separations, and both the characteristics and outcomes of the heterogeneous separations. Importantly, different from their work, we examine the underlying sources of differential earnings losses, including those attributable to destination employer pay premia and the nature of worker–employer matches formed post-separation. This analysis also contributes to studies of the sources of earnings losses and the roles of employer-specific pay premia and worker sorting (Lachowska, Mas, and Woodbury 2020; Fackler, Mueller, and Stegmaier 2021; Schmieder, von Wachter, and Heining 2023) and the job ladder (Krolikowski 2017; Jarosch 2023). Overall, our paper underscores that the reason for separation, even within the same adverse event, conveys important information about post-displacement trajectories.

Beyond their empirical importance, our findings are relevant for quantitative models that take these estimates as key inputs to discipline parameters, analyze the mechanisms behind these losses, and evaluate policy (Huckfeldt 2022; Jarosch 2023; Braxton, Herkenhoff, and Phillips 2024).

The rest of the paper is organized as follows. Section 2 explains our data and methodology. Section 3 presents our main results. Section 4 concludes.

2 Data and empirical methodology

2.1 Canadian Employer-Employee Dynamics Database

We use the CEEDD between 2001 and 2016, an annual employer-employee matched, longitudinal administrative record of the universe of Canadian individual and employer income tax filings, with a separate form providing key job separation information. The CEEDD links *person*-level tax returns, *employer*-level National Accounts Longitudinal Microdata File (NALMF) records from employer tax returns, and *job*-level returns and ROE. On the worker side, the database provides information on demographics (e.g., age and gender) and earnings from *all* jobs. We define earnings as total *annual* pre-tax earnings (constant 2010 Canadian dollars) from employment. Like most administrative data used in the literature, ours do not contain data on hours worked.

In Canada, separating workers need the ROE form to provide authorities with information to assess eligibility for EI, as well as the benefit entitlements. The ROE contains information on employer and worker identifiers, exact hiring and separation dates, and the reason behind a job separation.[1] By law, employers are required to issue an ROE whenever there is an "interruption in earnings," which can be caused by termination of work or a large decline in regular earnings. Importantly, changes in business ownership (e.g. M&As) can generate changes in employer identifiers without interrupting earnings. When earnings are uninterrupted and the new employer has access to the payroll records, an ROE need not be issued. Appendix A provides additional details about the data, institutional details, and the ROE form.

2.2 Separations within a mass layoff

Our sample selection criteria are chosen in a manner consistent with previous work (Jacobson, LaLonde, and Sullivan 1993; Lachowska, Mas, and Woodbury 2020). This allows for a comparison with prior work before incorporating the ROE data. Accordingly, we focus on employers with at

[1]Reassuringly, employer EI contributions in Canada have not been experience-rated since 2001, removing a key financial incentive for employers to disguise layoffs as quits.

least 50 employees in every year from 2002 to 2007, positive employment in 2006 and 2007, and a non-missing industry code in 2007. We limit our sample to the working-age population between the ages of 20 and 50 in at least one year during 2008–2010 and long-tenure workers, defined as individuals who have been continuously employed by the same primary employer from 2002 through 2007. For workers with multiple jobs during the year, we define the primary employer as that which accounts for the highest share of earnings. We also restrict the sample to individuals with positive earnings in every year from 2002 through 2014 and for whom an employer identifier is available.[2] Thus, our estimates reflect the outcomes of highly-attached workers.

Following Lachowska et al. (2020), we define a *separator* as a long-tenure worker who is separated from her primary employer at any point during 2008–2010. A separator is classified as a *mass-layoff separator* if the separation occurs in a year during which her primary employer experiences a mass layoff.[3] For the years between 2008 and 2010, a mass-layoff event occurs when (i) an employer experiences an employment drop of 30% or more relative to its 2007 employment and (ii) its 2007 employment does not exceed 130% of 2006 employment. The second condition reduces the chances of classifying temporary employment fluctuations as mass layoffs (Davis and von Wachter, 2011). Finally, a *stayer* is a worker who remains attached to the same primary employer between 2002 and 2014. In our analysis, the treatment group consists of mass-layoff separators, while the control group incorporates all job stayers as in Lachowska et al. (2020).[4]

We adopt the job flow exclusion methodology in Benedetto et al. (2007) to filter out employer identifier changes associated with M&As, legal restructuring, and cross-establishment movements within a parent organization. A separation is excluded if (i) 80% or more of the origin employer's

[2]Further, we focus on the years 2002 to 2014 due to an oil price shock in 2015 in Canada that induced highly sector-specific job separations.

[3]The group we label *mass-layoff separators* tend to be what the literature labels as *displaced workers*, defined as workers who experience an involuntary job separation. We retain this distinction throughout this paper as the ROE reveals that not all mass-layoff separators are laid-off workers.

[4]Appendix B.4 considers alternative control groups, and Figure A2 in Appendix B.2 examines how the composition of separations changes under alternative employment-contraction thresholds.

Table 1: Mass-layoff separations and sample characteristics

(a) Separation composition

ROE reason	Share (%)	Share among ROEs (%)	Outflow (%)	Inflow (%)
Layoff	24.0	46.8	14.4	14.4
Quit	9.9	19.3	2.2	5.1
Other	17.4	33.9	22.3	22.8
Missing	48.7	–	58.3	51.7

(b) Worker and employer characteristics

	Mass-layoff separators					Stayers
	Average	Layoff	Quit	Other	Missing	
Worker characteristics						
Average earnings 2002–2005 (2010 CAD)	53,546	52,996	54,873	56,357	46,967	56,457
Log average earnings, net of employer premium	10.573	10.541	10.649	10.643	10.716	10.665
Female (proportion)	0.313	0.282	0.386	0.398	0.505	0.522
Age in 2007 (years)	39.45	40.34	37.29	39.22	39.89	40.74
	(6.77)	(6.54)	(6.84)	(6.91)	(6.94)	(6.17)
Fraction received EI	0.64	0.77	0.32	0.46	0.09	0.16
Average EI among recipients (2010 CAD)	12,542	13,212	8,573	10,594	9,466	8,602
Employer characteristics in 2007						
Employer size (number of workers)	3,997	2,397	7,862	5,966	25,150	9,575
	(10,338)	(4,486)	(17,185)	(14,961)	(28,911)	(22,469)
One-digit NAICS industry (proportions)						
1 agriculture, forestry, fishing	0.020	0.024	0.009	0.006	0.009	0.003
2 mining, utilities, construction	0.038	0.037	0.040	0.019	0.024	0.040
3 manufacturing	0.634	0.720	0.424	0.413	0.141	0.189
4 trade, transportation	0.122	0.081	0.221	0.302	0.606	0.159
5 information, finance, professional services	0.125	0.086	0.220	0.156	0.101	0.126
6 educational and health care services	0.014	0.011	0.021	0.023	0.017	0.364
7 arts, recreation, hospitality services	0.035	0.035	0.034	0.059	0.087	0.019
8 other services	0.005	0.003	0.008	0.013	0.004	0.015
9 public administration and unclassified	0.008	0.002	0.023	0.008	0.011	0.085
Number of employers (pre- and post-separation)	21,250	15,510	8,860	10,675	10,920	12,825
Number of workers	21,465	15,180	6,285	11,015	30,850	774,075

Note: Panel (a) reports separation reasons for the conventional mass-layoff separator sample. Shares among ROEs are conditional on a recorded ROE; missing denotes employer-identifier changes without an ROE. Outflow and inflow measure concentrated worker flows as defined in the text. Panel (b) describes layoffs, quits, other recorded reasons, and employer-identifier changes without ROEs during mass layoffs in 2008–2010, as well as stayers attached to the same primary employer from 2002–2014. Log average earnings net of the firm premium is a measure of worker quality. EI statistics cover the separation year and the following year for separators and 2008–2009 for stayers. The "Average" column pools layoffs and quits. Parentheses contain standard deviations. Note that the number of employers row for the "Average" column need not be the sum of "Layoff" and "Quit" columns, as there are some employer overlaps. Counts are rounded to the nearest 5 due to data confidentiality.

total workforce moves to the same destination (concentrated outflow), or (ii) 80% or more of the destination employer's total employees are new hires from the same origin (concentrated inflow).

Even after excluding such concentrated flows, Panel (a) of Table 1 shows that 48.7% of mass-layoff separations in our sample are in fact employer identifier changes without an ROE. These

employer identifier changes can arise when a business changes ownership while workers remain employed without an interruption in earnings. Under the conditions described above, no ROE is required. However, a change in identifier is labeled as a separation by conventional approaches.

We argue that employer identifier changes without an ROE are more likely to be spurious separations (not actual job losses). We provide three additional pieces of evidence to support this claim. First, the last two columns of Panel (a) show that separations without an ROE have a significantly higher likelihood of being associated with concentrated flows (e.g., employer merger, acquisition, and reorganization activities). The values in the column "Outflow" are constructed as follows. For a given mass-layoff separator moving from employer A to B, we compute the ratio of the total number of individuals who move from employer A to B to the total employees at employer A. We report the average of these ratios across separators for each job separation reason. The "Inflow" column instead uses the number of employees at employer B as the denominator. Among mass-layoff separators without an ROE, an average of 58.3% of all employees from the origin employer move to the same destination employer. Further, on average, workers originating from the same employer as the separator represent 51.7% of all employees at the destination employer.[5]

Second, there is a negligible change in average earnings for mass-layoff separators with a missing ROE after 2008, similar to earnings paths of stayers (Figure A1 in Appendix B.1). Third, only a small fraction (less than 10%) of such separators receives EI (Table 1 Panel (b)). Taken together, the preponderance of evidence suggests that missing-ROE separations are likely spurious, although we cannot rule out the possibility that some involve actual separations.

Methodologically, our results show that conventional approaches used to exclude concentrated flows from mass-layoff samples, as in Benedetto et al. (2007) and implemented in Section 2.2, do not eliminate all such flows. Section 2.3 provides guidance for researchers using data without supplemental information like the ROE, who must instead rely solely on employer identifier changes.

[5]While the degree of flow concentration is high among missing-ROE separations, it need not be 100%. First, M&As may not involve the entire organization. Second, ROEs may be missing for reasons unrelated to flow concentration (e.g., independent contractors or non-compliance).

Panel (a) shows that, of all mass-layoff separations, 24.0% are recorded as layoffs, accounting for 46.8% of ROE separations. Thus, more than half of mass-layoff separations with ROE are actually due to quits or other reasons. This finding implies that, even when focusing on ROE separations, identifying involuntary job losses using separations during large employer contractions, i.e., "mass layoffs," as in previous studies, captures many separations that cannot readily be classified as involuntary. This result complements the findings in Flaaen et al. (2019), who show that 45% of mass-layoff separations are reported as arising for reasons other than firm distress.

For the remainder of the paper, we largely focus on separations recorded as layoffs or quits. We do so for two reasons. First, the evidence above suggests that missing-ROE observations often do not represent actual job losses. Second, we also find that separations that we group under "Other" in Table 1 are less likely to be actual separations. Around 80% of separations in this group are coded under "code K", a category that includes administrative changes that need not involve an interruption in earnings (see Appendix A). Other major categories in this group cover idiosyncratic reasons (e.g., pregnancy, illness, schooling, and retirement), which we do not intend to differentiate.

Next, we compare the characteristics of workers who are laid off or quit during mass layoffs with those of stayers. Panel (b) of Table 1 reports their demographic and labor market characteristics, with the "Average" column pooling layoffs and quits.

We document key differences across groups. First, relative to stayers, laid-off workers had around 3,500 CAD lower average earnings (2002–2005) and lower log average earnings net of the employer premium (proxying for worker quality), while those who quit had more similar earnings along both dimensions.[6] Second, laid-off workers are more likely to be male and older than workers who quit. Third, laid-off workers are more likely to receive EI transfers and to receive larger amounts than those who quit. Fourth, the primary employers of laid-off workers have a smaller number of workers than those of workers who quit. Fifth, layoffs are highly concentrated within manufacturing, which accounts for 72% of layoffs, consistent with a sharp decline in the industry during the Great Recession. Quits are also prevalent in trade and transportation and in information, finance, and

[6]Figure A13 Panel (b) also shows that workers who quit tend to be higher-type workers.

professional services. Finally, those with missing ROEs tend to originate from larger employers and from wholesale, retail, and transportation, consistent with the fact that reorganizations leading to employer identifier changes are more likely to occur in larger organizations.

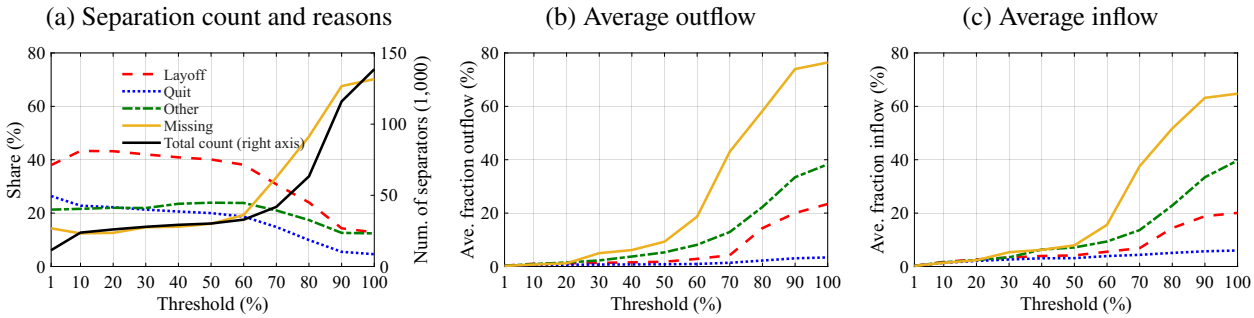
The differences in EI receipt warrant some discussion. In our sample, 32% of mass-layoff quits and 16% of stayers receive EI. There are reasons why EI receipt need not imply a layoff. First, while workers who quit without just cause are ineligible for EI, those who have just cause (e.g., harassment) may qualify. Second, even if workers stay with their employer, they may receive EI if they experience non-employment spells due to seasonal work.

2.3 Identifying concentrated flows without the ROE

Most strategies developed to identify involuntary separations in employer–employee matched datasets infer separations from employer identifier changes. To account for restructuring or mergers, these studies commonly filter out concentrated worker flows between employers. As discussed above, an oft-used approach follows Benedetto et al. (2007) and excludes separations with concentrated inflows or outflows of at least 80%. Variants of this procedure exist (see Hethey-Maier and Schmieder 2013; Halla et al. 2020), but the underlying principle of tracking concentrated flows is the same. These approaches share a common trade-off: lowering the threshold filters out more spurious separations but also reduces sample size and excludes genuine cases. Our data allow us to evaluate this trade-off directly, since the ROE records the reason for separation.

Panel (a) of Figure 1 shows the distribution of separations (i.e., employer identifier changes) after dropping concentrated flows that exceed each threshold. A notable pattern is the decline in separations with missing ROEs until around the 50% threshold, after which the share stabilizes. Clearly, a lower threshold also implies a smaller sample of separations, as shown by the black line. However, this decline is clearly driven by the removal of separations with missing ROEs. Panels (b) and (c) show that fractions of concentrated outflows and inflows—as defined previously—for reported layoffs and quits are small even at the most relaxed threshold of 100%. The concentration also drops significantly up to the 50% threshold, especially for those with missing ROEs.

Figure 1: The impact of varying threshold for concentrated flows



Note: For each threshold, Panel (a) presents the distribution of separations after concentrated flows that surpass the threshold are dropped (left axis) and the corresponding total number of separations remaining (right axis, solid black line). Panel (b) presents the fraction of the total workforce of an employer who exit and transition into the same employer, averaged over each mass-layoff separator (outflow). Panel (c) presents the ratio of new hires who originate from the same employer to the total workforce of the destination employer, averaged over each mass-layoff separator (inflow).

We note that the 50% threshold is a useful illustration and benchmark for this trade-off in the context of the CEEDD data. However, a cutoff that generates similar desirable properties in other datasets may depend on data- and country-specific features which include the prevalence of partial acquisitions, as well as conventions of tracking employer identifiers.

2.4 Estimating the consequences of job separation

We now discuss our methodology to estimate earnings losses upon separations during mass layoffs. We then explain how we use the AKM and Woodcock (2015) estimators to evaluate the importance of employer and match effects in driving these earnings losses.

To estimate the scarring effects of job separation on earnings, we follow Jacobson et al. (1993) and Lachowska et al. (2020) and use a distributed lag regression:

$$y_{i,t} = \alpha_i + \zeta_t + \beta x_{i,t} + \sum_{s \in S} \sum_{k=-4}^{6} d^s_{i,t,k} \times \gamma^s_k + \varepsilon_{i,t}. \tag{1}$$

Here, $y_{i,t}$ denotes the log of annual earnings of individual i reported in year t, while α_i and ζ_t capture individual and time fixed effects, respectively. Further, $x_{i,t}$ is a vector of individual and primary employer characteristics including a quadratic in age, gender and age interactions, and interactions between year dummies and workers' earnings (averaged over 2002–2005), employer size (2007), and one-digit industry (NAICS) code (2007). The vector of dummy variables $d^s_{i,t,k}$ indicates that the worker i in year t is observed k years before, on, or after a separation. For each mass layoff separator, we assume that separation happens a year prior to an employer identifier change for

that individual (i.e., $k = 1$ denotes the first year at the new employer). We estimate the effects of separation for all mass-layoff separators ($s = \text{all}$) and separately for $s \in \{\text{layoffs}, \text{quits}\}$, with stayers being the control group. Under the assumption that, absent separation, average earnings of mass-layoff separators would be parallel to those of stayers, estimated γ_k^s is interpreted as the causal effect of separations. We cluster standard errors by worker and current employer to accommodate (i) correlation over time within workers and across workers employed by the same employer and (ii) first-stage estimation error in employer premia common to workers within the same employer.

The changes in earnings upon job separation can then be decomposed into three main components as in Lachowska et al. (2020): those that arise from changes in employer-specific pay premium, match effect, and a residual effect.

We identify employer-specific time-invariant effects on earnings, termed "employer-specific pay premium," using the following Abowd, Kramarz, and Margolis (1999) (AKM) regression:

$$y_{i,t} = \kappa_i + \psi_{j(i,t)} + \lambda_t + v_{i,t}. \tag{2}$$

We regress the log of annual earnings $y_{i,t}$ on individual fixed effects κ_i, year fixed effects λ_t, and employer fixed effects $\psi_{j(i,t)}$, where $j(i,t)$ denotes individual i's primary employer in year t. For this regression, following the literature, we use a different sample, restricting the full database of earnings between 2001 and 2016 to exclude (i) observations belonging to stayers and mass-layoff separators used in estimating Equation (1), (ii) earnings in the first or last year of an employment spell, (iii) earnings below 400 times the national average minimum hourly wage, and (iv) employers with fewer than 5 employees in that year. In the end, we have a sample of 92,352,170 worker-years.[7]

Following the interpretation of Card et al. (2013), $\widehat{\psi}_j$ represents a measure of pay advantages associated with being employed by a particular employer j. We assign $\widehat{\psi}_j$ to each worker-year observation whenever possible and use them as outcomes of job separations in Equation (1).

We estimate match effects as time-invariant worker–employer fixed effects following Woodcock (2015) and Lachowska et al. (2020). We first compute average log earnings (net of year effects)

[7]We re-estimate AKM premia when we include stayers and separators in the sample (Figure A11), and provide additional results using AKM and BLM estimates (Figures A12 and A13).

$\overline{y}_{ij}$ for each worker–employer pair (i, j) over the match. We then regress these average earnings on worker θ_i and employer ξ_j fixed effects, weighting by match duration. Formally,

$$\overline{y}_{ij} = \theta_i + \xi_j + \mu_{ij}. \tag{3}$$

The estimated match effects $\widehat{\mu}_{ij}$ capture the component of earnings attributable to time-invariant match effects, averaged over the match duration, after controlling for estimated worker and employer effects θ_i and ξ_j, respectively. To estimate Equation (3), we use the same AKM sample described above, except that we keep stayers and separators. We then estimate the impact of separations on match effects using $\widehat{\mu}_{ij}$ as the dependent variable in Equation (1).

To allow for direct comparisons, we first follow the approach of Lachowska et al. (2020) in estimating employer and match effects using the AKM and Woodcock estimators, respectively. We then address concerns about the AKM's assumption of exogenous mobility by reporting estimates derived from the approach in Bonhomme et al. (2019) (BLM). To do so, we consider two different models, one with additive worker and employer effects and another where both interact. In the additive model, employer pay premiums are restricted to be common across worker types, while the interactive model allows employer pay premiums to vary by worker type. Section 3 compares AKM with BLM results, and Appendix C describes these methods and their assumptions.

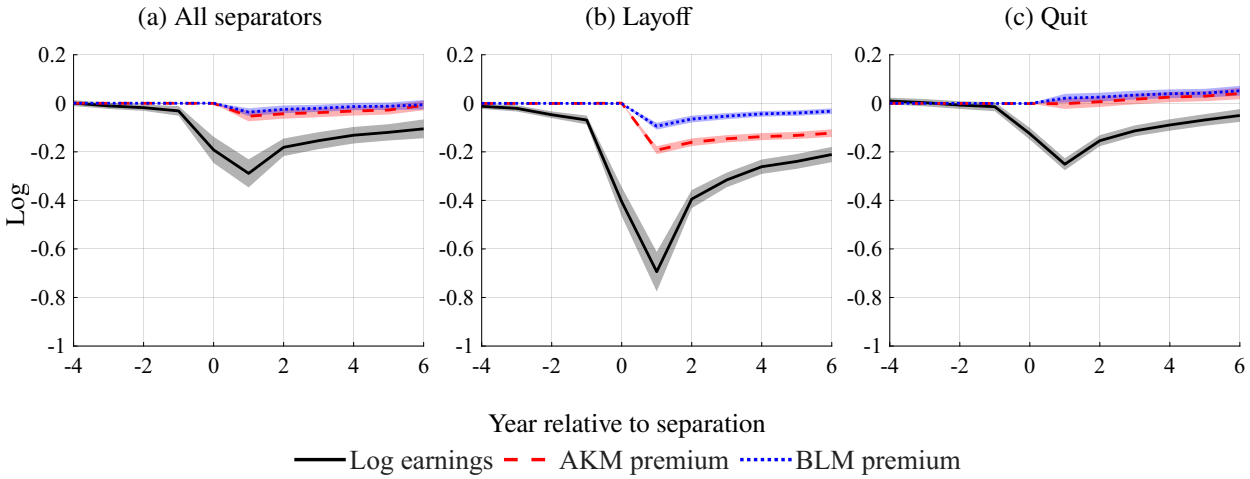
3 Earnings losses and the role of employers

In this section, we first document earnings and employer effects outcomes for *all* mass-layoff separations, providing a benchmark to prior estimates. We then show that both the size and the sources of earnings losses differ sharply between separation types within mass layoffs.

3.1 Pooled estimates versus layoffs and quits

We start by estimating the dynamics of log earnings and employer-specific pay premiums for *all* mass-layoff separations. The objective here is to establish a benchmark consistent with the literature by retaining the same approach to mass layoff identification and intentionally *excluding* ROE information. Thus, we first analyze earnings losses for all mass-layoff separators, including

Figure 2: Earnings and employer-premium dynamics following mass-layoff separations



Note: This figure plots estimates for earnings and employer-specific pay premium dynamics upon job separation by reason of separation during mass layoffs. Panels (a), (b), and (c) present estimates for all mass-layoff separators, layoffs, and quits, respectively. All separators include layoffs, quits, other reasons, and missing ROEs. Solid-black lines show estimated γ_k^s for log earnings from Equation (1), while dashed-red and dotted-blue lines represent estimates for employer-specific pay premiums using AKM and additive BLM models, respectively. Shaded areas in the corresponding colors show 95% confidence intervals.

layoffs, quits, other reasons, and missing ROEs. For this sample, Panel (a) of Figure 2 shows that the average earnings loss in the year following the separation was 25% (29 log points) during the 2008–2010 episode. Importantly, the earnings of separators remain 10% lower even after six years past the job separation. These results fall within the range of existing estimates from admin data in the U.S. and Europe, between 18% and 46% in the year of separation and between 10% and 30% five years after (Jacobson et al. 1993; Couch and Placzek 2010; Davis and von Wachter 2011; Lachowska et al. 2020; Schmieder et al. 2023).

Having established benchmark estimates, we now examine the extent to which this average masks differential scarring effects of displacements between layoffs and quits. Panels (b) and (c) provide the earnings dynamics of workers who are laid off or who quit during mass layoffs in 2008–2010, respectively. One year post-separation, laid-off workers experience an average earnings loss of 69 log points, while those who quit experience a loss of 25 log points. Six years post-separation, earnings remain 21 log points lower for layoffs, but only 5 log points lower for quits.[8]

[8]That quitters still experience earnings losses relative to stayers does not imply that quitting was worse than remaining at their distressed employer. For example, workers may accept lower earnings to avoid an anticipated deterioration of their employment prospects in their existing jobs.

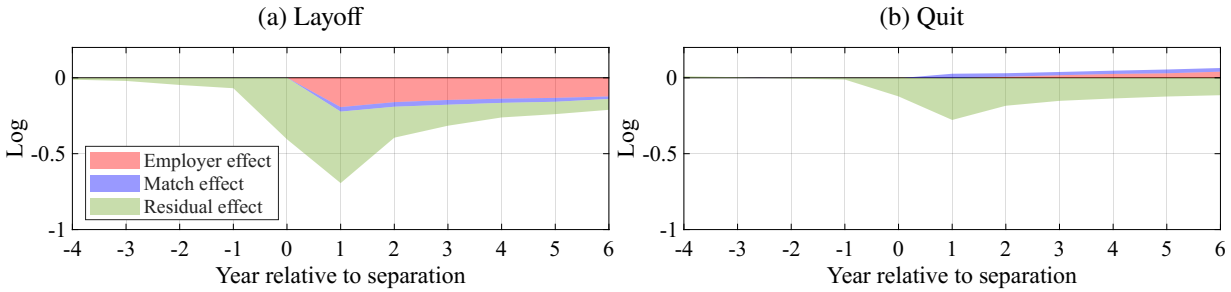
We note that *even if* we retain only layoffs and quits to obtain a sample free from spurious and idiosyncratic separations—an optimistic scenario for databases without ROE data—the resulting estimates still understate layoff losses by about one-sixth in the first year and one-quarter after six years. When we pool layoffs and quits, short- and long-term losses are 57 and 16 log points, compared with 69 and 21 for layoffs (Figure A3 in Appendix B.3). Thus, the understatement reflects both the inclusion of spurious separations and the mixing of layoffs with quits.

Next, we show that accounting for the reason for separation is also relevant for quantifying the role of employer-specific pay premia in explaining earnings losses upon separations. We find that ignoring the reason for job separation by mixing all separations leads to a severe understatement of the role of employer-specific premia in explaining earnings losses. In particular, for all separations, the employer-specific premium is 5 log points lower in the year following the separation and 1 log point lower six years after, explaining only around 18% of short-term and 8% of long-term losses.

Once we account for the reason for job separation, the dynamics of employer-specific pay premium are drastically different for layoffs and quits in mass layoffs. While laid-off workers face significant and long-lasting losses in employer-specific premium, those who quit see *no* short-term decline and even a long-term *gain*, as shown by the red lines in Figure 2. For laid-off workers, the average employer-specific premium is 19 log points lower in the year following the layoff and 12 log points lower six years after. These imply that the loss of employer-specific premium accounts for 28% of earnings losses in the short term and 58% in the long term. In contrast, for quits, the employer premium remains unchanged in the first year and rises by 4 log points after six years.

Finally, Figure 2 includes the dynamics of additive BLM estimates (blue lines). BLM estimates decrease by 9 log points in the first year for layoffs, while AKM estimates drop by 19 log points. This divergence persists even in the long run, with the BLM premium remaining at 3 log points lower for layoffs, and 5 log points higher for quits. Despite these level differences, the layoff-quit gap in employer premium dynamics is substantial in either specification.

Figure 3: Decomposing the sources of earnings losses upon mass-layoff separation



Note: This figure presents a decomposition of earnings losses upon job separation by reason of separation, as indicated in the ROE, from employers experiencing a mass layoff. Earnings losses are decomposed into those attributable to changes in AKM employer premiums, match effects, and residual effects. The residual is the log-earnings coefficient less the employer-premium and match-effect coefficients.

3.2 Employer, match, and residual components

Recall from Section 2 that we estimate the dynamics of employer premia and match effects upon mass-layoff separations. This analysis allows us to decompose earnings losses into components attributable to employer premia, match effects, and a residual effect unexplained by either.

Figure 3 presents the results. For quits, on average, AKM employer effects and match effects are both positive and thus mitigate the decline in earnings due to a decline in residual effects. Meanwhile, for layoffs, employer effects are negative and large, match effects are negative but small, while residual effects decline sharply, especially in the short run. For layoffs (quits), employer, match, and residual effects are -19 (0), -3 (3), and -47 (-28) log points relative to stayers in the year following the separation. The large short-run residual effects may reflect periods of non-employment or pay reductions in a new job arising from, say, the loss of tenure-related pay not captured by employer and match effects, as discussed by Lachowska et al. (2020).

A key insight from Figure 3 is that the *long-run* earnings loss gap between layoffs and quits is largely attributable to the disproportionately large loss of employer effects among layoffs. In particular, the total earnings loss gap six years post-separation is 16 log points (21 versus 5), while the corresponding gap in employer premia is also 16 log points (-12 versus 4). Thus, net of employer premia, the long-run earnings gap largely disappears. While the residual losses play an important role in explaining short-run losses for both groups, the differences in employer premia account for the entire long-run gap in average earnings losses between layoffs and quits.

Table 2: Below-, on-, and above-diagonal sums and averages: AKM and BLM

	AKM–Woodcock Employer-premium quintiles			BLM Employer types		
	Below diagonal	On diagonal	Above diagonal	Below diagonal	On diagonal	Above diagonal
(a) Layoff						
Share of separators	0.488	0.366	0.144	0.371	0.346	0.283
Average change in log earnings	-0.295	-0.024	0.173	-0.348	-0.049	0.059
Average change in employer effect	-0.374	-0.004	0.306	-0.309	0.000	0.255
Average change in match effect	0.023	-0.047	-0.169	0.024	0.000	-0.045
Average residual effect	0.057	0.027	0.036	-0.064	-0.049	-0.151
(b) Quit						
Share of separators	0.281	0.380	0.339	0.313	0.265	0.422
Average change in log earnings	-0.100	0.128	0.284	-0.091	0.144	0.245
Average change in employer effect	-0.366	0.015	0.361	-0.273	0.000	0.306
Average change in match effect	0.194	0.038	-0.128	0.068	0.000	-0.048
Average residual effect	0.072	0.075	0.051	0.115	0.144	-0.013
(c) Average						
Share of separators	0.428	0.370	0.201	0.354	0.323	0.323
Average change in log earnings	-0.258	0.022	0.228	-0.282	-0.003	0.130
Average change in employer effect	-0.373	0.002	0.333	-0.300	0.000	0.274
Average change in match effect	0.056	-0.021	-0.149	0.036	0.000	-0.046
Average residual effect	0.060	0.041	0.043	-0.018	-0.003	-0.099

Note: This table presents five rows for separations by reason (layoff, quit, or an average of both) with below-, on-, and above-diagonal transitions: (i) the fraction of separators, (ii) average change in log earnings, (iii) average change in employer effects, (iv) average change in match effects, and (v) average residual effects of the transition. In the *AKM–Woodcock* block, below (above)-diagonal transitions represent moves to an employer in a lower (higher) employer-premium quintile, and on-diagonal transitions are moves to a same-quintile employer. Here, employer quintiles are determined by AKM effects. In the *BLM* block, transitions are moves to lower-type, same-type, or higher-type employers. The employer effect uses additive BLM premiums. The match effect is the interactive BLM change *net* of the additive employer component. See Appendix C for more details. In both blocks, the residual is the earnings change less the employer and match components. Employer and match effects are compared one year before and three years after separation; changes in log earnings use the log of average annual earnings over 2002–2005 as the pre-separation baseline. Shares may not sum to exactly one because of disclosure rounding.

That employer premia play such an important role in determining long-term earnings losses and differential outcomes between layoffs and quits warrants a deeper investigation into the reasons behind this divergence. To do so, we subdivide separators into different segments of the employer premium ladder. Specifically, we assign each employer to a quintile based on its AKM estimate. Each separation is then assigned to one of 25 quintile-to-quintile transitions based on the separator's origin and destination employer quintiles. We then compare pre-separation earnings (2002–2005 average), match effects, and employer effects with their values three years after separation.

In Table 2, below-diagonal transitions represent moves to a destination employer in a lower employer-effects quintile; on-diagonal and above-diagonal transitions represent moves to a same-

quintile employer and to a higher-quintile employer, respectively.[9] We note three key patterns.

First, laid-off workers (Panel (a)) are more likely than quitters (Panel (b)) to transition into lower quintiles. Almost half of those laid-off move to lower-paying employers, while less than 30% do so for quits. Further, Figure A4 shows that laid-off workers are more likely to originate from high-paying employers and face a greater probability of moving down from their origin quintile.

Second, conditional on falling into lower employer-premium quintiles, workers who are laid-off incur larger declines in earnings than those who quit (29.5 vs. 10.0 log points). Importantly, employer premium losses are comparable for layoffs and quits with a below-diagonal transition (37.4 vs. 36.6 log points) and the smaller loss in earnings for quits is mostly driven by a larger gain in match effects (2.3 vs. 19.4 log points), mitigating lost employer effects. This suggests that some workers who quit accept lower employer premiums but are compensated with better matches.

Finally, conditional on rising along employer-premium quintiles, quits experience larger increases in earnings relative to layoffs (28.4 vs. 17.3 log points). Above-diagonal transitions show substantial gains in employer effects and losses in match effects, with employer premium gains slightly larger and match effect losses slightly smaller for quits than layoffs.

We repeat the same analysis using BLM estimates but now group employers by their BLM types instead. The employer effect uses additive BLM premiums, while the match effect is the interactive BLM change *net* of the additive employer component (see Appendix C for more details). Table 2 shows that layoffs are still more likely to make downward moves than quits (37.1 vs. 31.3%), while quits are more likely to move up (42.2 vs. 28.3%). For downward moves, quitters still experience much smaller earnings losses and larger match gains, but the gap in match effect gains narrows.

3.3 Discussion

In this section, we revisit our main results under alternative samples, mass-layoff identification strategies, and empirical specifications, with details outlined in Appendix B.4.

[9] These estimates may not match those in Figures 2 and 3, since they are descriptive changes rather than regression estimates relative to stayers.

First, we examine whether our results rely on the earnings measure or restrictions on positive earnings. The layoff–quit gap in earnings losses remains when measuring earnings in levels (instead of in logs) and when including workers with zero earnings (Figure A5). Further, excluding low earners reduces the size of earnings losses, but retains large differences between layoffs and quits.

Second, we study how outcomes vary with tenure and industry. Earnings losses are more severe for long-tenure workers, with employer-premium losses playing a large role in explaining differential outcomes by tenure (Figure A6). While manufacturing workers incur worse outcomes, losses remain substantially more severe for layoffs than for quits even in non-manufacturing (Figure A7), implying that industry composition alone cannot explain the gap between layoffs and quits.

Third, we consider alternative samples and mass-layoff definitions. The main conclusions remain similar when we exclude multiple-job holders, follow the mass-layoff identification and sample selection of Davis and von Wachter (2011), or restrict attention to employer closures (Figure A8). In the employer closure case, earnings losses are relatively smaller for layoffs and larger for quits, but the gap remains substantial. In addition, lowering the concentrated-flow exclusion threshold to 50% produces much larger losses for all separators, while earnings and employer-premium losses for layoffs and quits remain similar to the baseline estimates (Figure A9).

Finally, we examine whether our findings depend on control group choice or differences in individuals' earnings trajectories. We allow stayers to include those who separate post-2010, compare separators and stayers within the same employer, and allow for worker-specific time trends (Figure A10). In all cases, the gaps in outcomes between layoffs and quits remain large.

4 Conclusion

This paper sheds light on the complex anatomy of mass layoffs using Canadian employer-employee matched administrative data combined with detailed job separation records. We document stark differences in earnings and employer pay premiums between workers who are laid off and those who quit, with layoffs experiencing more severe and persistent losses. Importantly, the differences in post-separation employer-premium losses account for much of this long-term earnings-loss gap.

These differences, even within the same mass layoff, show that separations are not uniform. Workers at a distressed employer make different labor market decisions, experience varied labor market outcomes, and potentially respond differently to labor market policy changes. Our findings underscore the importance of modeling distinct types of separations in quantitative analyses of job dissolution and policy evaluation. For example, payroll subsidies could discourage employers from laying off long-tenure workers. Advance notice requirements may help workers avoid large earnings losses by giving them time to transition to another employer.

Beyond the heterogeneity based on the reason for separation studied in this paper, the ROE also allows for an analysis based on the timing of separations in mass layoffs and how they are shaped by worker and employer decisions. Employer distress may trigger early quits, while other workers may choose to quit after the mass layoff to generate better offers. Employers may also decide which workers to lay off and in what sequence. We leave these considerations for future research.

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Online Appendix

A Data

In this section, we provide additional details on our database.

The Canadian Employer-Employee Dynamics Database (CEEDD) is an employer-employee matched database which covers the universe of individual and corporate tax filers in Canada. It is a linked collection of administrative data from various government agencies maintained by Statistics Canada. This paper utilizes information from a subset of forms used to construct the CEEDD:

- Individual-level tax files: This is obtained from the T1 Income Tax and Benefit Return, the main tax return used by individuals to file annual income taxes. It consolidates information on income earned from all sources, including those derived from employment, businesses, and investments. It also contains detailed demographic and financial information.

- Employer-level tax files: The National Accounts Longitudinal Microdata File (NALMF) combines tax and administrative forms submitted by employers, including the T2 Corporation Income Tax Return, T4, Payroll Account Deductions (PD7), and Goods and Services tax/Harmonized Sales tax (GST/HST). Any enterprise that files at least one of these forms is included in the NALMF. Thus, the NALMF includes all corporate tax filers and unincorporated businesses with at least one employee but excludes non-employer businesses.

- Job-level tax files: Employers must issue a T4 Statement of Remuneration Paid for each employee. The T4 contains information on various forms of compensation, including wages and salaries, tips or gratuities, bonuses, taxable benefits, and commissions. Individuals who received compensation from multiple employers during the year would have multiple T4s. We also use the Record of Employment (ROE) form, discussed in more detail below.

Each T1 form features an individual identifier, while each NALMF record features an employer identifier. The T4 job-level records contain *both* individual and employer identifiers and allow for linkages between T1 individual demographics and financials with NALMF employer characteristics.

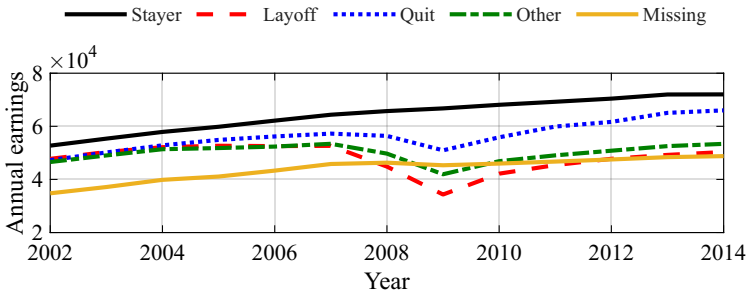
Record of Employment (ROE). As discussed in Section 2.1, the ROE is a form that employers must issue to employees whenever an interruption in earnings occurs. An interruption in earnings occurs when at least one of the following two conditions is met. First, an employee experiences seven consecutive calendar days with no work and no insurable earnings from the employer. This condition covers separations associated with a layoff, quit, or termination. Second, the employee's salary falls below 60% of their regular weekly earnings and the interruption is caused by reasons such as illness, injury, maternity/parental leave, child care among others.

The ROE contains data on employer and worker identifiers, hiring and separation dates, and the reason for a job separation. This information is available only when an ROE is issued.[1] Key to this study is the information on the reason for a job separation. The ROE includes codes for shortage of work (layoff), strike or lockout, schooling, illness or injury, quit, maternity, retirement, work-sharing, apprentice training, dismissal or suspension, leave of absence, parental leave, compassionate/family care, and "Code K." The "Code K" category includes technical cases involving a change in payroll or ownership, a change in company name, or a change in pay period.

Institutional details. In Canada, separating workers need the ROE form to provide authorities with information to assess eligibility for employment insurance (EI), as well as the benefit amount and duration they are entitled to. These workers are ineligible for regular EI benefits and severance payments based on years of service if they quit the job without just cause. As such, a potential concern is that employers may have incentives to disguise or misreport a layoff as a quit on the ROE. This would be especially concerning if the EI program featured experience rating as in the U.S., where an employer's payroll tax rate would rise with the number of previous employees receiving benefits. Facing such consequences, employers possess incentives to reduce the likelihood that their ex-employees' EI applications succeed (Lachowska, Sorkin, and Woodbury, 2025). In the context of submitting the ROE, disguising a layoff as a quit may prevent a worker from receiving EI.

[1] When an individual finds a new job after separating, the new job's start date is observed only if that employer later issues an ROE. Non-employment duration can therefore be calculated only for workers who subsequently separate from the new job.

Figure A1: Average annual earnings: Separators by ROE reason



Note: This figure shows average annual earnings over time for stayers and mass-layoff separators whose employer identifier changes between 2008 and 2009. Separators are grouped by ROE reason, including those with missing ROEs. Stayers are shown by the solid black line. Earnings are denominated in 2010 CAD.

Reassuringly, unlike the U.S. system, employer EI contributions in Canada are not experience-rated based on an employer's layoff history. As such, an employer has less of a motive to misreport a layoff as a quit. Misreporting in the other direction (i.e., disguising a quit as a layoff) is also less likely as employers who do so become liable for paying severance packages. A final safeguard against misreporting is that eligibility for other transfers (e.g., disability, parental leave, and retirement) for former employees is contingent on the reason for job separation. Since the ROE has significant implications for various worker outcomes, employees can appeal or challenge incorrect entries and are protected by employment laws. In fact, misrepresentation of information on ROEs faces severe financial penalties, both for claimants and employers. While some degree of non-compliance and misreporting is inevitable (e.g., under-the-table arrangements), the information provided in the CEEDD, merged with ROE data, is collected within a favorable institutional framework.

B Additional results

We provide additional results to supplement our discussions in the main text.

B.1 Outcomes of separators with missing ROE

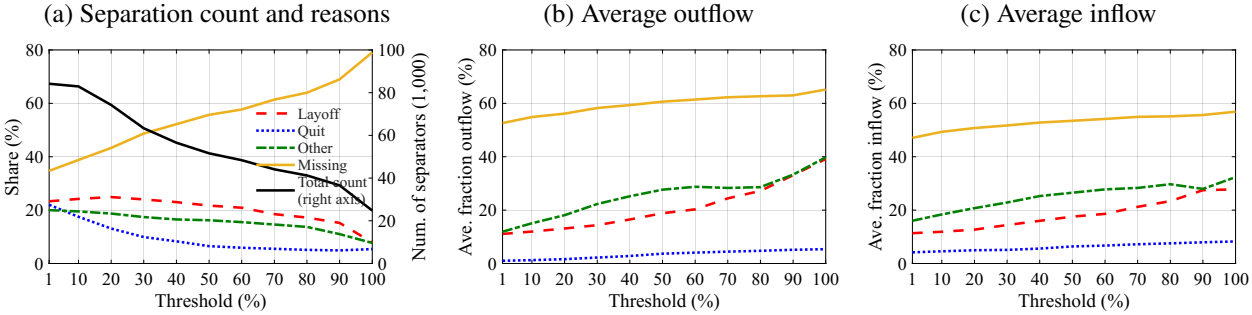
We explore the earnings dynamics of mass-layoff separators with missing ROEs, as discussed in Section 2.2. Figure A1 presents average annual earnings over time for stayers and mass-layoff separators whose employer identifier changes between 2008 and 2009. The earnings of workers who are laid off, who quit, and who separate for other reasons decline in 2009. In contrast, the earnings of missing-ROE separations are similar to those of stayers, with no comparable decline.

B.2 Mass-layoff definition: alternative employment-contraction thresholds

In Section 2.2, we outline our approach to identifying mass-layoff events. As a baseline, we follow Lachowska et al. (2020) and define a mass layoff during 2008–2010 as requiring, among other criteria, that an employer experience an employment reduction of at least 30% relative to its 2007 level. While this threshold is common in the literature, its choice may influence the types of mass-layoff events—and associated separations—captured by the identification strategy. For example, a low employment-contraction threshold may capture employers that are not truly undergoing mass layoffs, thereby introducing noise, while a high threshold may restrict the sample to cases that more closely resemble plant closures or M&As, with the latter manifesting as widespread changes to employer identifiers in the data. To examine this point, Figure A2 presents summary statistics for the sample of mass-layoff separations obtained under alternative employment-contraction thresholds. For each threshold, the growth of employment between 2006 and 2007 is also restricted to be less than the corresponding percentage while all other identification criteria (including the 80% exclusion threshold for concentrated flows) are held fixed.

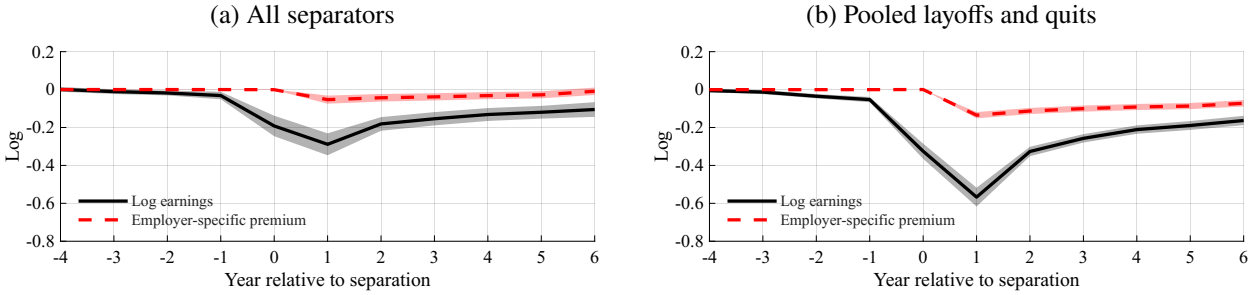
We highlight three observations. First, the share of layoffs among all identified separations is close to its maximum at the 30% threshold (Panel (a)). Second, increasing the employment-contraction threshold results in a higher proportion of missing-ROE separations (Panel (a)) due to a greater incidence of concentrated-flow events associated with large employment contractions, as suggested by Panels (b) and (c). Finally, raising this threshold not only leads to a substantial reduction in the number of identified separators (Panel (a)), but also to a remaining sample that is both small and increasingly composed of workers with missing ROEs and ties to concentrated-flow events. Taken together, these findings suggest that the 30% threshold strikes a reasonable balance: It effectively captures events primarily driven by layoffs, and—when combined with the concentrated-flow exclusion criterion used by Benedetto et al. (2007)—it also helps reduce the incidence of spurious separations associated with missing ROEs.

Figure A2: Effects of varying employment-contraction threshold for mass-layoff identification



Note: This figure presents various statistics associated with varying the employment-contraction threshold used for identifying a mass-layoff event. For each threshold, employment growth between 2006 and 2007 is also restricted to be less than the corresponding percentage. For each threshold tested, Panel (a) presents the distribution of all identified separations (left axis) and the corresponding total number of separations remaining (right axis, solid black line). Panel (b) presents the fraction of the total workforce of an employer who exit and transition into the same destination employer, averaged over each mass-layoff separator (outflow). Panel (c) presents the ratio of new hires who originate from the same employer to the total workforce of the destination employer, averaged over each mass-layoff separator (inflow).

Figure A3: Earnings and employer-premium dynamics: Pooled layoffs and quits



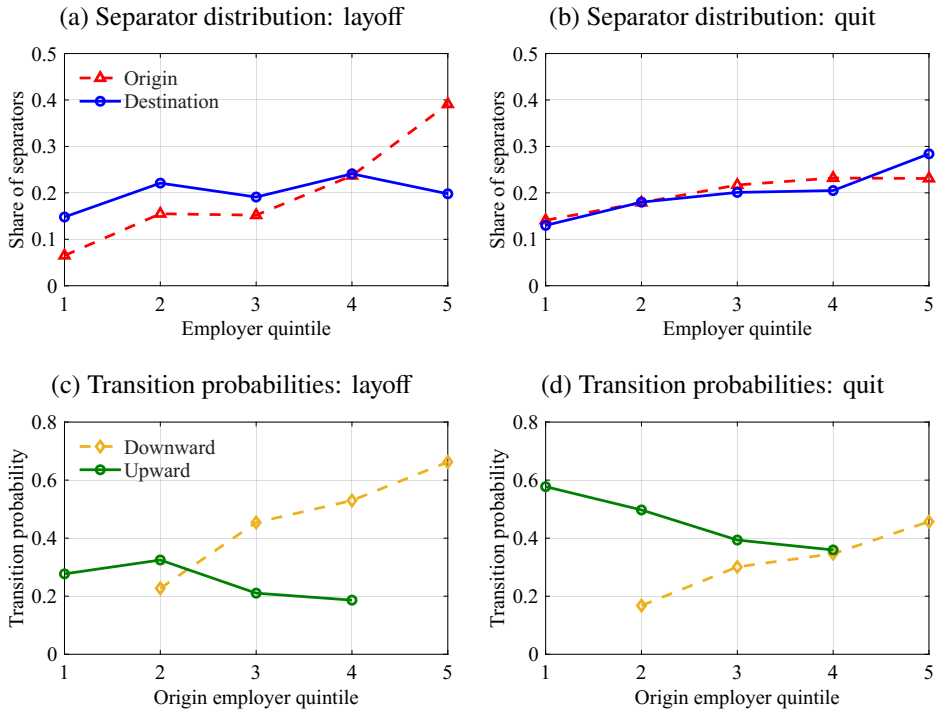
Note: This figure plots estimates for earnings and employer-specific pay premium losses upon job separation during mass layoffs. Panel (a) repeats the baseline, which includes all separators. Panel (b) pools layoffs and quits only. Solid-black lines show estimated γ_k^s values for earnings losses and dashed-red lines show estimated γ_k^s values for employer-specific pay premium losses in Equation (1). Shaded areas in the corresponding colors show 95% confidence intervals.

B.3 Pooled outcomes and transitions across the employer effect distribution

This section presents results which supplement the findings in Section 3. Figure A3 presents the earnings and employer-premium dynamics when we pool only layoffs and quits (Panel (b)), alongside those of all mass-layoff separators (Panel (a), replicated from Panel (a) of Figure 2). Panel (b) offers an approximation of pooled outcomes from a database that is not affected by spurious employer identifier changes or separations for other idiosyncratic reasons. Here, the first-year and sixth-year earnings losses rise from 29 to 57 log points and 11 to 16 log points, respectively. Employer premium losses also become more severe. Nevertheless, pooling layoffs and quits still severely understates the earnings losses of layoffs, as shown by Panel (b) of Figure 2.

Next, we further examine the asymmetries in employer premium dynamics between layoffs and

Figure A4: Transitions across employer effect distribution



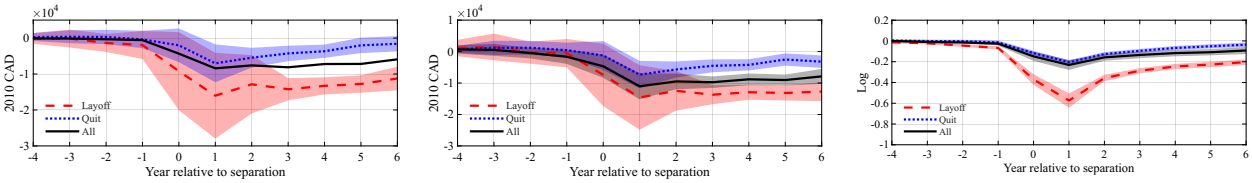
Note: Panels (a) and (b) plot the distribution of separations by origin (dashed-red) and destination (solid-blue) employer effect quintiles for layoffs and quits in mass layoffs, respectively. Panels (c) and (d) present upward (solid-green) and downward (dashed-orange) transition probabilities by origin employer effects quintiles for layoffs and quits, respectively. Values are based on a comparison of outcomes between one year before and three years after separation.

quits in Figure A4. Panels (a) and (b) present the distribution of separations by origin (dashed-red) and destination (solid-blue) quintiles for both types of separations. Comparing origin-quintile distributions, laid-off workers are more likely to originate from high-paying employers, whereas quits have a more even distribution. For example, 39% of layoffs during mass layoffs originate from top-quintile employers, while only 7% come from the bottom quintile. These shares are 23% and 14% for quits, indicating much less heterogeneity along the origin-quintile distribution. Further, the share of layoffs that head to the top destination quintile is only 20%, while for quits it is 28%.

Panels (c) and (d) present downward and upward transition probabilities by origin employer quintiles for layoffs and quits in mass layoffs, respectively. Comparing layoffs and quits, we note two key takeaways. First, downward transition probabilities are larger for laid-off workers, especially when the origin employer is in the fourth and fifth employer-effect quintiles. Upward transition probabilities, on the other hand, are larger for workers who quit. Second, downward transition probabilities increase more with the origin-employer quintile for layoffs than for quits.

Figure A5: Earnings dynamics under alternative earnings definitions and samples

(a) Level changes in earnings (b) Including zero earnings (c) Excluding low earners



Note: This figure plots estimates for earnings losses upon job separation by reason of separation during mass layoffs. Panel (a) reports losses in levels (2010 CAD) for the baseline sample. Panel (b) reports losses in levels (2010 CAD) when the sample is modified to allow for individuals who do not report positive earnings. Panel (c) reports log earnings losses when the sample is modified to exclude individuals with very low earnings. Specifically, worker observations reporting annual earnings below 400 times the national average minimum hourly wage—the same threshold employed in the AKM sample—are excluded. Dashed-red and dotted-blue lines (along with shaded 95% confidence intervals) show estimated earnings losses for layoffs and quits, respectively, while the solid-black line presents these losses for all mass-layoff separators.

B.4 Results under alternative specifications and samples

In this section, we repeat our main result in Figure 2 of Section 3.1 under alternative samples, mass-layoff identification, and empirical specification, as summarized in Section 3.3.

Earnings losses in levels. First, we consider earnings losses in levels (instead of in logs) for our baseline sample. Panel (a) of Figure A5 shows earnings losses (in 2010 CAD) by reason for separation during mass layoffs. For layoffs (quits), earnings decline by $\$16,000$ ($\$7,000$) in the year following the separation and stay $\$11,200$ ($\$1,600$) lower six years after.

Including zero earners. The baseline sample requires that workers report positive earnings for the entire duration considered (2002–2014). Next, we relax this restriction by allowing for workers with zero earnings in certain years and therefore report results in levels. Results for this alternative sample in Panel (b) of Figure A5 are similar to those in Panel (a), which uses the baseline sample.

We argue that our baseline assumption is more reasonable because tax filers with zero earnings for an entire year are rare, so the positive earnings requirement primarily excludes non-filers. Relaxing this restriction adds workers with missing data and leads to an unbalanced panel, especially toward the end of the sample. Further, in this unbalanced panel, we see a declining fraction of workers with reported earnings over time, possibly due to emigration. Thus, including individuals with zero or missing earnings introduces compositional changes. To address emigration, we conduct an additional analysis that includes only workers with missing earnings who returned by the end of the sample. Since far fewer individuals are added back into the sample, the results remain similar.

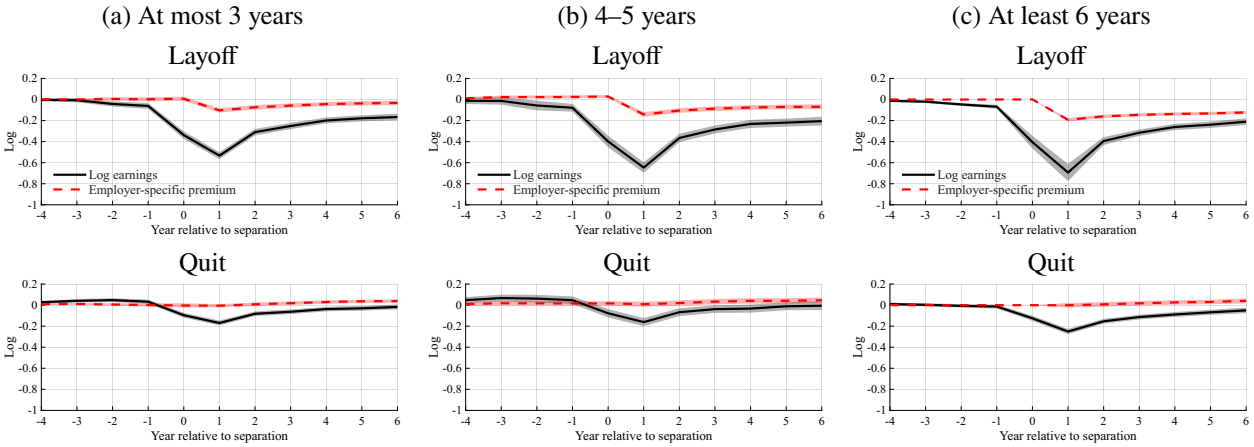
Excluding low earners. Returning to log earnings, we now analyze whether very low earnings are a primary driver of the estimated losses. The use of log earnings as our main outcome variable in Equation (1) may imply that very low earnings will translate to very large (log) earnings losses, which should not be interpreted directly as percentage declines. To ameliorate concerns around this issue, we re-run the earnings loss regression on a sample that excludes low-earners, defined as observations with annual earnings below 400 times the national average minimum hourly wage (the same threshold for the AKM sample). Panel (c) of Figure A5 shows that this reduces the estimated magnitude of earnings losses for layoffs by around 12 log points. Nevertheless, the central finding—substantial heterogeneity in outcomes between layoffs and quits—remains robust.

Heterogeneity by tenure. We also analyze how earnings and employer premium losses vary with tenure among separators. Panels (a), (b), and (c) of Figure A6 present outcomes for workers with at most three, four to five, and at least six years of tenure (baseline), respectively. For Panels (a) and (b), we *do not* require workers to be attached to the same employer six years prior to separation.

We find that earnings losses are more severe for long-tenure workers. One explanation is that high-tenure workers experience a greater loss of employer-specific, non-transferable human capital. Including short-tenure workers, who have accumulated less employer-specific capital, reduces the magnitude of losses. In addition, we find that losses in employer pay premiums play a large role in explaining differences in earnings losses among laid-off workers. This means that, apart from the loss of employer-specific human capital, long-tenure workers also tend to move to lower-paying employers. This could arise if higher-paying employers retain workers better, thus leading to higher concentrations of well-paid separators who find re-employment with lower-paying employers.

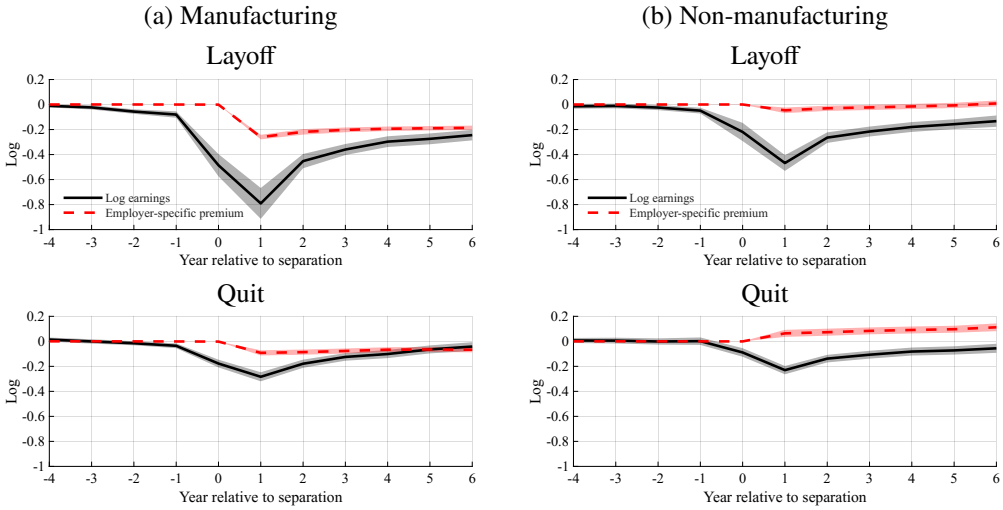
Manufacturing and non-manufacturing. In order to analyze whether the differential outcomes between layoffs and quits arise due to differences in their industrial composition, we present earnings and employer-premium dynamics separately for separators from manufacturing and non-manufacturing industries in Figure A7. While outcomes are worse for manufacturing workers for both layoffs and quits, earnings losses are much more severe for layoffs than for quits in both sectors.

Figure A6: Earnings and employer-premium dynamics by tenure



Note: This figure plots estimates for earnings and employer-specific pay premium losses upon job separation by reason of separation during mass layoffs. Tenure is first measured as of 2007. Panels (a), (b), and (c) present results for workers with at most three, four to five, and at least six years of tenure, respectively. For columns (a) and (b), the baseline requirement that workers be continuously employed by the same employer for at least six years is relaxed. Column (c) repeats the baseline results. Layoffs (quits) are shown in the top (bottom) row. Solid-black lines show log earnings and dashed-red lines show employer premiums. Shaded areas in the corresponding colors show 95% confidence intervals.

Figure A7: Earnings and employer-premium dynamics by sector

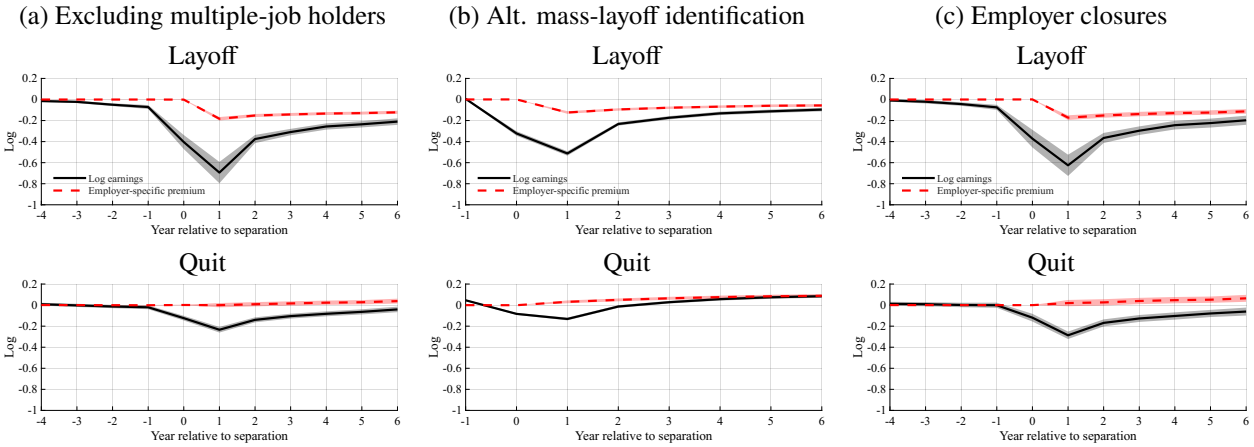


Note: This figure plots estimates for earnings and employer-specific pay premium losses upon job separation by reason of separation during mass layoffs. Layoffs (quits) are shown in the top (bottom) row. Solid-black lines show estimated γ_k^s values for earnings losses and dashed-red lines show estimated γ_k^s values for employer-specific pay premium losses in Equation (1). Shaded areas in the corresponding colors show 95% confidence intervals. Panel (a) presents results for manufacturing and Panel (b) for non-manufacturing.

Excluding multiple-job holders. Our main sample includes workers with multiple jobs in a year, with the primary employer defined as the one with the largest share of annual earnings. Panel (a) of Figure A8 shows that, even after excluding workers with multiple jobs, the earnings and employer effect dynamics for layoffs and quits remain similar to the baseline results shown in Figure 2.

Implementing an alternative mass-layoff identification. Next, we evaluate the removal of the restriction that separations occur within 2008–2010 and explore alternative sample restrictions and

Figure A8: Earnings and employer-premium dynamics under alternative samples



Note: This figure plots estimates for earnings and employer-specific pay premium losses upon job separation by reason of separation during mass layoffs. Layoffs (quits) are shown in the top (bottom) row. Solid-black lines show estimated γ_k^s values for earnings losses and dashed-red lines show estimated γ_k^s values for employer-specific pay premium losses in Equation (1). Shaded areas in the corresponding colors show 95% confidence intervals. In Panel (a), we drop individuals with multiple job records. In Panel (b), without restricting separations to be within the 2008–2010 period, we consider an alternative sample and adopt definitions of long-tenure workers and mass-layoff events following Davis and von Wachter (2011). In Panel (c), we focus only on employer closures when identifying mass-layoff events in our sample.

definitions of long-tenure workers and mass-layoff events, following Davis and von Wachter (2011). Consider a reference year t. A mass layoff occurs in year t when: (i) employment drops more than 30% between $t-2$ and t, (ii) employment in $t-2$ is not more than 130% of employment in $t-3$ and (iii) employment in $t+1$ is less than 90% of employment in $t-2$. We focus on individuals who are at most 50 years old and who have been employed with the same primary employer for at least three years, as in Davis and von Wachter (2011). Similar to them, we define a mass-layoff separator in year t to be an individual who separates from their primary employer in year t, while the employer is identified as having experienced a mass layoff in year t or $t+1$.[2] We also restrict attention to all observations two years before and six years after the separation year t. For any given year t, this implies a panel with at most nine observations per worker, with the earliest observation for all workers being at $t-2$. To maintain this restriction, we are left with reference years t from 2003 to 2010, given our data. For any given nine-year window for a reference year t, a stayer is defined as an individual who maintains positive earnings with the same primary employer from $t-2$ to $t+6$. Finally, employer effects are estimated repeatedly for each reference year t as in Section 2.4, where the exclusion of stayers and separators is specific to the sample in each t.

Using the 2001 to 2016 data, we identify mass layoffs, mass-layoff separators, and stayers for

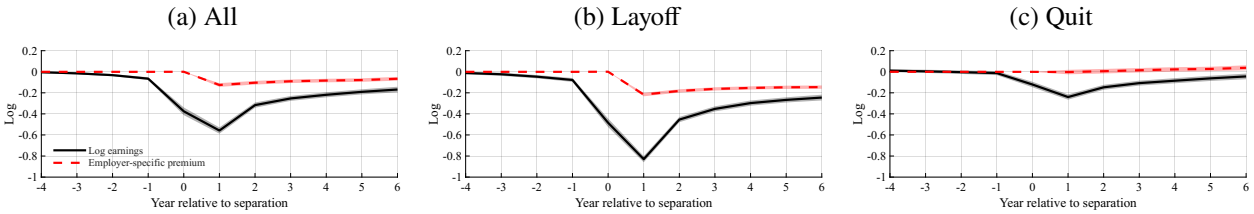
[2]We also drop separations associated with concentrated flows, as in Section 2.2.

each reference year t from 2003 to 2010. We then estimate the regression in Equation (1) using pooled data from each panel constructed using each reference year t. As in Davis and von Wachter (2011), our controls include an interaction of year dummies with a worker's average earnings (over $t-2$ to t). Panel (b) of Figure A8 shows that earnings and employer effects losses are only slightly smaller for both layoffs and quits when compared with Figure 2.[3] Importantly, our main conclusions also remain similar: Gaps in earnings and employer effects losses between layoffs and quits are still large and employer effects account for earnings losses for layoffs but not for quits.

Focusing on employer closures. As an alternative to identifying mass-layoff events from large employment contractions, we consider the subset of mass-layoff separations occurring between 2008 and 2010, as defined in Section 2.2, that are associated with employer closures. Formally, these closures are defined as employers that experience a mass layoff and either register zero employment or drop out from the sample for some given year, without ever returning to employment until the end of the sample period. Given our long-tenure sample restriction that workers must be attached to the same primary employer from 2002 through 2007, no plant closures occur before 2008. To illustrate, a mass-layoff separation that occurs in 2009 from an employer that eventually disappears permanently from 2013 onward would be considered a separation associated with an employer closure. The results in this case are presented in Panel (c) of Figure A8. Restricting attention to employer closures retains the key message that laid-off workers suffer much larger earnings losses than quitting workers. However, compared with our baseline estimates, we note that earnings losses are slightly lower for the layoffs that are associated with an employer closure (63 vs. 69 log points) and slightly higher for quits (29 vs. 25 log points). This result can be rationalized by the fact that mass layoffs may involve some discretion by employers in terms of selecting who to lay off and by workers in terms of deciding when to quit to join a new employer. An employer closure dilutes the negative selection for layoffs, as argued by Gibbons and Katz (1991), and may also reduce the strategic opportunities for workers to time their quits.

[3]The shorter pre-displacement horizon is due to the lower three-year tenure requirement.

Figure A9: Outcomes under stricter exclusion threshold for concentrated flows

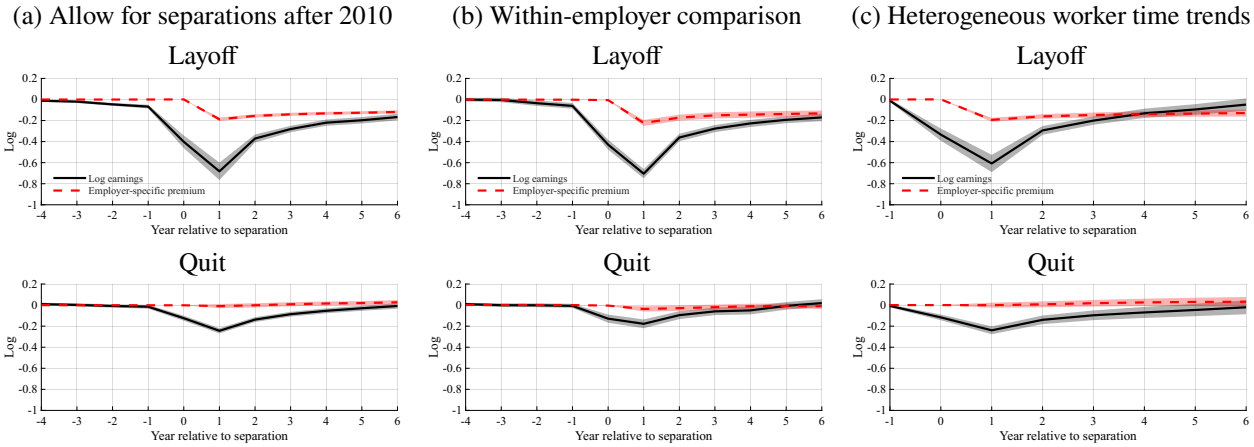


Note: This figure plots estimates for earnings and employer-specific pay premium losses upon job separation by reason of separation during mass layoffs. Panel (a) presents estimates for all mass-layoff separations, Panel (b) presents estimates for those due to layoff, and Panel (c) presents estimates for those due to quits. Solid-black lines show estimated γ_k^s values for earnings losses and dashed-red lines show estimated γ_k^s values for employer-specific pay premium losses in Equation (1). Shaded areas in the corresponding colors show 95% confidence intervals. Here, we lower the exclusion threshold associated with concentrated flows from 80% to 50%.

Alternative exclusion threshold. As outlined in Section 2.2, we adopt the job flow exclusion methodology employed by Benedetto et al. (2007) to partially filter out employer identifier changes associated with various forms of firm reorganization. Section 2.3 shows that while the standard exclusion threshold mitigates the inclusion of missing ROEs into our mass-layoff separator sample, significant reductions of employer identifier changes of this sort can be achieved by a 50% threshold without reducing the sample of other valid separations much further. Panel (a) of Figure A9 shows that for all mass-layoff separators, the earnings decline is significantly larger in this case compared to the baseline in Figure 2 (56 versus 29 log points). This larger loss is attributed to the stricter threshold, which reduces the presence of missing ROEs. A sample with missing ROEs dampens earnings and employer premium losses, so sample restrictions that are better at excluding spurious separations (in the absence of ROE information) result in larger earnings losses. Moreover, Panels (b) and (c) indicate that earnings and employer premium losses for layoffs and quits are broadly similar to those in Figure 2. This provides reassurance that the stricter exclusion threshold effectively targets spurious separations without significantly altering the sample of layoffs and quits.

The main estimates presented in Section 3 depend on whether stayers' earnings growth provides a satisfactory comparison for the counterfactual earnings growth of separators had they not separated. For example, stayers may take on additional roles after a mass layoff and experience higher earnings growth. In the following discussions, we examine alternative control groups and earnings trends. First, we relax the restriction that stayers remain with the employer throughout the whole sample to reduce the possibility of selecting stayers with exceptionally stable employment and earnings paths

Figure A10: Earnings and employer-premium dynamics under alternative control groups and trends



Note: This figure plots estimates for earnings and employer-specific pay premium losses upon job separation by reason of separation during mass layoffs. Each panel shows layoffs in the top row and quits in the bottom row. Solid-black lines show estimated γ_k^s values for earnings losses, and dashed-red lines present estimated γ_k^s values for employer-specific pay premium losses. Shaded areas in the corresponding colors show 95% confidence intervals. In Panel (a), we expand the control group to include workers who, while not experiencing an employer separation between 2008 and 2010, did separate from an employer in the period following 2010, using Equation (1). In Panel (b), we compare outcomes between mass-layoff separators and stayers from the same employer, using Equation (A1). In Panel (c), we allow for heterogeneous (worker-specific) trends, using Equation (A2), a version of Equation (1).

(and those who are more likely to take on additional roles in an employer, if any). Second, we compare stayers and separators within the same employer and allow for worker-specific earnings trends to account for any differences in earnings growth trajectories between separators and stayers.

Alternative control group that allows for separations after 2010. Our baseline approach follows Lachowska et al. (2020) and sets the control group to include only stayers who never separate from their original employer throughout the sample. A concern with this approach arises from Bertheau et al. (2023), who demonstrate that restricting the control group to workers who remain continuously with the same employer overstates the estimated effects of job displacement. To address this point, we expand the control group to include workers who did not separate from their employer during the 2008–2010 period (the mass-layoff window), but who may have separated thereafter. As shown in Panel (a) of Figure A10, this broader control group leads to a roughly one log point reduction in year 1 earnings losses among laid-off workers, but ultimately yields similar estimates of the earnings impact of job loss, and does not meaningfully alter the gap between layoffs and quits.

Comparing separators and within-employer stayers. When estimating the effects of mass-layoff separations, we compare the outcomes of long-tenure mass-layoff separators with those of long-tenure workers who stay employed. We now redefine the control group to be the colleagues

of mass-layoff separators who remain with their employers. As such, we compare separators and stayers from the same employer using the following specification, as in Jacobson et al. (1993):

$$y_{i,t} = \alpha_i + \zeta_t + \eta_{j_0(i),t} + \beta x_{i,t} + \sum_{s \in S} \sum_{k=-4}^{6} d^s_{i,t,k} \times \gamma^s_k + \varepsilon_{i,t}. \tag{A1}$$

The difference from Equation (1) is that we include differential year effects by initial employer, $\eta_{j_0(i),t}$, in addition to year effects, ζ_t, where $j_0(i)$ denotes worker i's employer in 2002. Thus, we compare separators and stayers from the same initial employer, controlling for earnings changes common to workers originating from that employer. When stayers at an employer experiencing a mass layoff also systematically face earnings losses, the magnitude of estimated γ^s_k's in Equation (A1) may be smaller than those estimated in Equation (1).

Panel (b) of Figure A10 shows that earnings and employer premium losses are slightly less severe in this case for both layoffs and quits. However, gaps in earnings and employer effect losses between layoffs and quits remain very similar to those in Figure 2.

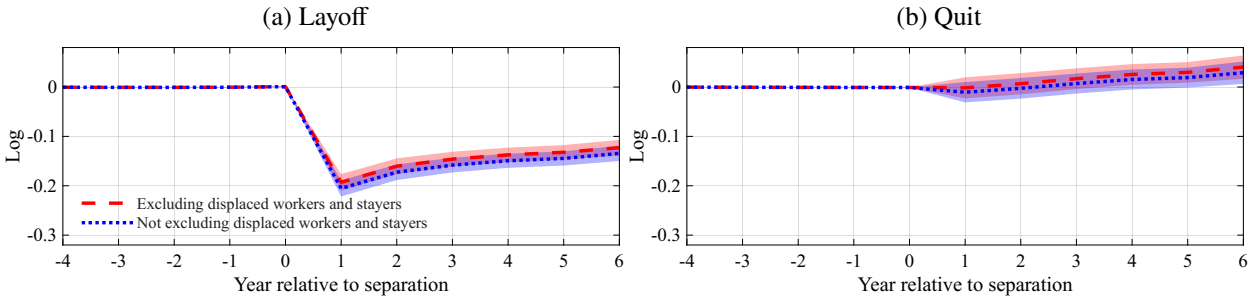
Incorporating heterogeneous time trends. Estimates from the main specification in Equation (1) may be biased if there are worker fixed effects in earnings growth. For example, if those with lower lifetime earnings growth are more likely to be laid off, estimated earnings losses may be overstated. To address this, we consider a specification with worker-specific linear time trends:

$$y_{i,t} = \alpha_i + \xi_i t + \zeta_t + \beta x_{i,t} + \sum_{s \in S} \sum_{k=-1}^{6} d^s_{i,t,k} \times \gamma^s_k + \varepsilon_{i,t}, \tag{A2}$$

where ξ_i is the worker fixed effect in growth. Notice that we do not estimate the effects of separation for two, three, and four years before the separation. This is to ensure that the worker-specific linear time trends are well identified based on these three additional years of observations. With the inclusion of worker-specific trends, the estimates of γ^s_k now reflect the effect of a mass-layoff separation relative to stayers, controlling for differences in workers' unobserved characteristics that lead to differences in level as well as growth of worker outcomes.

Panel (c) of Figure A10 shows that the magnitudes of earnings and employer effect losses remain similar to those in Figure 2. This provides reassurance that differential trends in earnings growth do *not* significantly bias the estimated effects of mass-layoff separations in our baseline specification.

Figure A11: Effects of mass-layoff separations on AKM employer-specific premium



Note: This figure presents estimates of employer-specific pay premium losses when the employer premia are calculated using a sample that includes both the stayers and separators. Panels (a) and (b) compare the path of employer premium losses under the baseline AKM sample (red-dashed lines) with the expanded sample (blue-dotted lines), for layoffs and quits, respectively. Shaded regions represent 95% confidence intervals.

Including the displaced worker sample for AKM estimation. As outlined in Section 2.4, we estimate employer-specific premia using a sample that excludes stayers and mass-layoff separators. This restriction helps avoid a mechanical relationship between employer effects and the earnings losses of mass-layoff separators, which could otherwise overstate the impact of employer effects. To assess the implications of estimating AKM employer premia with a different sample and then using those estimates to decompose the relative earnings losses of displaced workers excluded from the initial estimation, we recalculate the dynamics of employer-specific premium losses using a full sample that includes both stayers and mass-layoff separators.

Figure A11 shows that using the full sample in the analysis results in slightly greater losses in employer-specific premia for layoffs and smaller gains for quits. This result suggests that the decline in employer premia is slightly larger when estimates are drawn from a more selected set of transitions, particularly those associated with mass layoffs. Thus, including displaced workers may exacerbate the endogenous mobility bias in the AKM estimation. Overall, we conclude that this sample change preserves our main conclusions in Figure 2.

C Addressing endogenous mobility bias

Throughout the paper, we have highlighted the role of employer-premium ladder dynamics in shaping disparities in outcomes for workers who separate during mass-layoff events. We now discuss certain limitations of AKM-based estimates of employer-specific premia. We then demonstrate the robustness of our results when applying corrective measures suggested in the literature.

Consider the AKM regression in Equation (2):

$$y_{i,t} = \kappa_i + \psi_{j(i,t)} + \lambda_t + \upsilon_{i,t},$$

where $y_{i,t}$ is log earnings of individual i in year $t = 1, 2, \ldots, T$, κ_i, ψ_j, and λ_t are fixed effects for worker, employer, and year, respectively, and $\upsilon_{i,t}$ is an error term that might include a time-varying match-specific component, satisfying $\mathbb{E}[\upsilon_t] = 0$. To simplify, we abstract from time effects λ_t, which can be achieved by subtracting year-specific averages from log earnings, assuming year-to-year changes in average log earnings do not reflect changes in worker or employer composition.

Since the employer effects are identified from movers, they are biased when mobility decisions are affected by the error term $\upsilon_{i,t}$. This will be the case, for example, when workers change employers after a negative shock to their match-specific component. To see this, consider workers moving from employer j in year t to employer j' in year t'. Their average earnings growth is

$$\underbrace{\mathbb{E}[y_{t'} - y_t | j_{t'} = j', j_t = j]}_{\text{earnings change for movers}} = \underbrace{\psi_{j'} - \psi_j}_{\text{change in employer premium}} + \underbrace{\mathbb{E}[\upsilon_{t'} - \upsilon_t | j_{t'} = j', j_t = j]}_{\text{endogenous mobility bias}}.$$

It is easy to see that the above equation identifies employer premium under the exogenous mobility assumption of AKM, which implies $\mathbb{E}[\upsilon_{t'} - \upsilon_t | j_{t'} = j', j_t = j] = 0$.

We outline several reasons endogenous mobility might arise (see Card, Heining, and Kline, 2013). First, sorting might occur based on idiosyncratic match effects. Especially for voluntary moves, workers are likely to move towards jobs with higher match effects, leading to an upward bias in the AKM estimator. In contrast, workers at employers undergoing a mass layoff may move to employers with lower match effects if they are laid off and must find a job quickly in a depressed labor market. Match effects may also reflect job-specific human capital that is destroyed upon separation, in which case the AKM estimator is biased downward. Second, time-varying worker effects may also predict transitions from one employer to another. This might reflect changes in workers' general skills or bargaining power. For example, under positive assortative matching, a worker may move to a better-paying employer after a positive shock to their general skill. Meanwhile, mass layoffs may weaken workers' bargaining power and bias estimated employer-premium changes downward.

Overall, the problem of endogenous mobility bias arises because the AKM framework does not

allow for worker mobility to depend on earnings realizations conditional on worker and employer heterogeneity. Bonhomme, Lamadon, and Manresa (2019) (BLM) proposed a method to address the endogenous mobility bias based on the assumption that log earnings and mobility decisions follow a joint first-order Markov process in a general model with time-varying interaction between worker and employer heterogeneity. The first-order Markov assumption (with additional mild assumptions) leads to a simple bias-correction procedure to the AKM estimator (which assumes additive and time-invariant worker and employer effects), as long as $\upsilon_{i,t}$ is not a permanent component.[4] This procedure allows us to address the endogenous mobility bias that originates from time-varying match or worker effects that are transitory in nature, but it does not allow the bias to solely reflect time-invariant match effects. To account for time-invariant match effects, a similar bias-correction method can be applied to the case of time-invariant interaction between worker and employer effects, a special case of the time-varying model of BLM. We provide our results from these two models in the following discussion.

C.1 Additive worker and employer effects

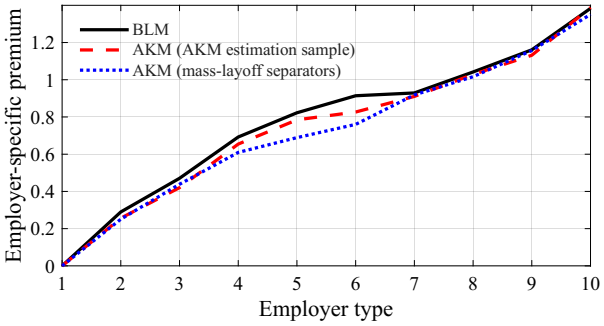
Here, we use the model that assumes additive worker and employer effects and run the same regressions as in Equation (1) to examine how the dynamics of employer premia upon a layoff and quit change. The BLM estimation strategy requires four periods of data, where workers only move employers between the second and the third period. We implement this procedure by constructing multiple subsamples of the AKM regression sample and pooling observations across subsamples.

Beyond the main results provided in Section 3, the comparison in Figure 2 also sheds new light on the differences between employer premia estimates from the AKM model and bias correction approaches like BLM. Our comparison shows that $\mathbb{E}[\Delta\hat{\psi}_{\text{AKM}}] < \mathbb{E}[\Delta\hat{\psi}_{\text{BLM}}]$ for displaced workers, particularly for layoffs. This difference suggests that, in our context, the AKM model underestimates employer pay premia associated with upward mobility (often linked to quits) and overestimates the losses from downward moves (often linked to layoffs). What might explain this disparity? Recall

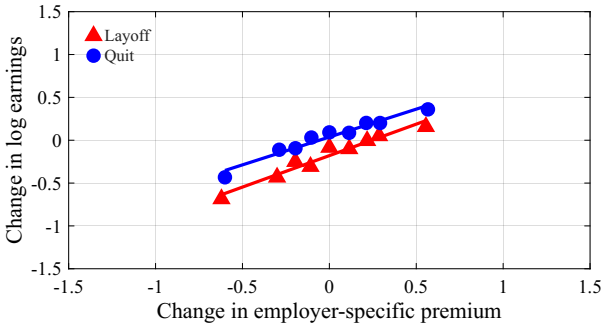
[4] A technical supplement with details on identification and estimation is available upon request.

Figure A12: Additional comparisons of AKM and BLM employer premia

(a) Employer-specific premium by employer type under AKM and BLM



(b) Changes in earnings vs. BLM employer-specific premium



Note: Panel (a) plots estimated employer-premium effects by employer type. The solid-black line shows employer-specific premia estimated with the procedure that largely follows BLM. For the red-dashed and blue-dotted lines, we obtain AKM employer premium averages for each employer type over each worker-year observation in two different samples. The red-dashed line represents averages for the AKM estimation sample (which excludes stayers and mass-layoff separators), while the blue-dotted line represents averages for mass-layoff separators (not including stayers). Panel (b) plots changes in log earnings for layoffs and quits when we divide each group of mass-layoff separators into nine groups based on the distribution of $\Delta\hat{\psi}_{\text{BLM}}$: quartiles for negative values (4 groups), zeros (1 group), and quartiles for positive values (4 groups). Employer premia are compared one year before and three years after separation, while earnings changes use average annual earnings in 2002–2005 as the pre-separation baseline.

that our analysis focuses on long-tenure workers separating from origin employers that underwent mass layoffs. Even when we exclude these workers from the AKM estimation, other workers displaced from these employers (e.g., low-tenure workers) remain in the AKM sample. If these workers experience larger earnings losses, say due to a decline in match effects, the AKM premium will be biased downward. The fact that the gap, as implied by $\mathbb{E}[\Delta\hat{\psi}_{\text{AKM}}] < \mathbb{E}[\Delta\hat{\psi}_{\text{BLM}}]$, is less pronounced for quits suggests that such workers who left these origin employers experienced a less severe drop in their match-specific component (e.g., possibly due to a stronger local labor market).

Additional results. Panel (a) of Figure A12 presents the estimated employer premia by employer type. The solid-black line is the premia estimated using the additive BLM approach. For the red-dashed and blue-dotted lines, we calculate the average AKM premia for each employer type (x-axis) across worker-year observations from two distinct samples. The red-dashed line shows averages for the AKM estimation sample, which excludes stayers and mass-layoff separators, while the blue-dotted line shows averages for only mass-layoff separators.[5] The estimated employer premia are similar in both the BLM estimates and the AKM estimates from the AKM sample, indicating that

[5]For the mass-layoff separator sample, we average employer premia at the worker-year level from 2002–2014, consistent with our sample. For the AKM sample, we use 2002–2015, the time period implied by utilizing the full dataset (2001–2016) but excluding the first and last year on the job.

endogenous mobility bias is not a significant concern in the AKM sample (which excludes mass-layoff separators). However, a more pronounced difference emerges when comparing the BLM estimates with the AKM averages for the mass-layoff separator sample. Specifically, mass-layoff separators tend to be associated with much lower employer-specific premia on average.

Panel (b) plots changes in log earnings for layoffs and quits when we divide each group of mass-layoff separators into nine bins based on the distribution of changes in BLM employer-premium estimates $\Delta\hat{\psi}_{\text{BLM}}$: negative-value quartiles (4 bins), zeros (1 bin), and positive-value quartiles (4 bins). This yields nine points in the scatter plot. We compare employer premia one year before and three years after separation, while earnings changes use average annual earnings in 2002–2005 as the pre-separation baseline. Consistent with Table 2, Panel (b) of Figure A12 shows that quits experience better earnings outcomes than layoffs for comparable changes in employer effects.

C.2 Interactive worker and employer effects

The bias-correction procedure for the model with additively separable worker and employer effects does not allow the bias to be entirely driven by time-invariant match effects. Now, we explicitly account for a time-invariant match-specific component in the log earnings equation. Here, there cannot be distinct worker or employer effects in addition to the match effect, because the match effect can be thought of as interactive worker and employer effects. Thus, we consider a model that only includes interactive worker and employer effects:

$$y_{i,t} = \phi_{l_i,k_{i,t}} + \upsilon_{i,t}, \tag{A3}$$

where $l_i \in \{1, 2, \ldots, L\}$ represents the discrete *type* of individual i, which entirely characterizes unobserved heterogeneity of the worker, $\phi_{l,k}$ is the effect of a match between worker type l and employer type k, and the error term satisfies $\mathbb{E}[\upsilon_t | l, k_t] = 0$.

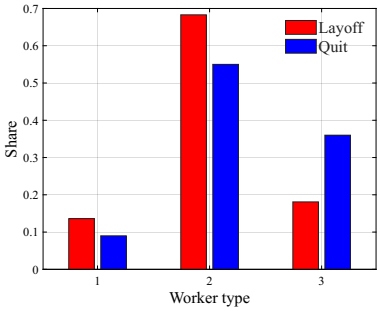
The interactive specification in Equation (A3) is more general and reduces to the additive specification in Equation (2) when $\phi_{l,k} = \kappa_l + \psi_k$. Note that the match effect $\phi_{l,k}$ is slightly different from that in the specification for the Woodcock estimator (μ_{ij}), which reflects the match effect *net* of worker and employer effects. However, it should be emphasized that the Woodcock estimator

Figure A13: Key results from estimation of time-invariant match effects

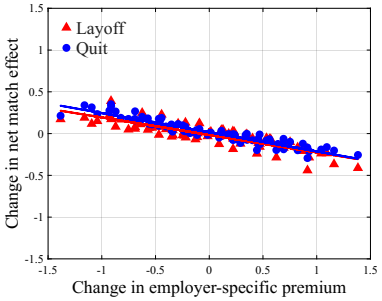
(a) Match effect ($\phi_{l,k}$) vs. employer premium (ψ_k)



(b) Distribution of unobserved worker types



(c) Δ net match effect ($\phi_{l,k} - \psi_k$) vs. employer premium (ψ_k)



Note: This figure plots the main results from the estimation of time-invariant match effects. Panel (a) shows the estimated match effect $\phi_{l,k}$ for each worker type $l \in \{1, 2, 3\}$ and for each employer type $k \in \{1, 2, \ldots, 10\}$. Panel (b) plots the distribution of separators across each worker type, by reason for separation. Panel (c) shows the average change in match effects net of employer effects $\phi_{l,k} - \psi_k$, where each dot represents an employer-type transition, for a total of 10×10 possible moves. These results are obtained by comparing outcomes between one year before and three after separation.

would be based on a mis-specified model if the specification in Equation (A3) is correct, and it does not address the endogenous mobility bias originating from the error term.

Notice that, with interactive worker and employer effects, there is no single employer premium that can be applied to all workers because changing employers has different effects across workers. Therefore, for our purposes, the interactive model must be estimated using our main sample of stayers and mass-layoff separators rather than a separate AKM sample. Here, we require that workers remain with the same employer for each pair of pre- and post-separation observations.

Figure A13 presents the key findings. Panel (a) shows the estimated match effect $\phi_{l,k}$ for each worker type $l \in \{1, 2, 3\}$ and employer type $k \in \{1, 2, \ldots, 10\}$. The estimated match effects are generally increasing in employer type, suggesting strong complementarity between employer and worker types. Panel (b) plots the distribution of separators across each worker type, by reason for separation. It shows that workers who quit are more likely to be higher-type workers than those who are laid-off. Finally, Panel (c) shows the average change in match effects net of employer effects $\phi_{l,k} - \psi_k$, where each dot represents an employer-type transition, for a total of 10×10 possible moves. These results are obtained by comparing outcomes between one year before and three after separation. We find that workers who experience larger declines in BLM employer-specific premia also experience larger gains in net match effects. Notably, the steeper slope for quits suggests that this relationship is somewhat more evident for quits than for layoffs, consistent with Table 2.